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Geographic and gender shape differences in the carapace of *Liocarcinus depurator* (Brachyura: Portunidae) using geometric morphometrics and the influence of a digitizing method

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Abstract

The current work describes the use of geometric morphometrics in the study of the portunid crab *Liocarcinus depurator* carapace shape variation along the Mediterranean coast of Spain. As a first step, the differences in shape induced by digitizing the carapace with either a digital camera or a flat bed scanner were investigated. Carapace shape information inferred from the camera images proved different from that obtained from the scanner images. This difference was greater than any shape variation because of crab gender or sample location. Carapace shape was analysed using multiple analysis of covariance (with centroid size as a covariate) of the partial warps and uniform components indicating differences between genders and between sample location; however, significant three-way interaction implied that such differences were gender and location specific. Landmarks identifying carapace width, carapace length and posterior carapace width showed greatest variation; hence gender and location differences were further analysed through the interrelationships between these linear measures. Regression analysis of the relationship between posterior carapace width and the ratio of carapace width to carapace length indicated clear differences between locations but not between genders. Crabs from the Alborán Sea had proportionally wider posterior carapaces than those from Alicante or North Catalonia. This finding may represent the difference between a southern influx of an Atlantic population distinguishable from a more northerly Mediterranean one, which would be in agreement with the overall oceanographic surface circulation pattern in the western Mediterranean.

Introduction

Liocarcinus depurator (L.) is the dominant brachyuran crab of Mediterranean demersal fisheries down to depths of around 200 m (Abelló, Valladares & Castellón, 1988; Abelló, Carbonell & Torres, 2002). It ranges in distribution from the coasts of the western Sahara to Norway, including the Mediterranean Sea (Udekem d'Acoz, 1999), where it is one of the most abundant species of continental shelf communities (Ungaro *et al*., 1999; Abelló *et al*., 2002). *Liocarcinus depurator* can be found on several types of substrata, although it is most commonly encountered on mud and sandy-muddy bottoms (Minervini, Giannotta & Falciai, 1982).

Morphometric studies on *L. depurator* have previously been performed, using linear measurements, in the western Mediterranean (Mori & Zunino, 1987; Abelló, Pertierra & Reid, 1990) and in the Atlantic coast of Spain (Fernández, González-Gurriarán & Freire, 1991). Fernández *et al*. (1991) found significant differences between the relationship width/length of *L. depurator* males and females in Ría de Arousa, with females being broader than males, whereas Abelló *et al*. (1990) did not observe any significant differences attributable to gender. Furthermore, the female maturity size of *L. depurator* was larger in Ría de Arousa than in the Mediterranean (Mori & Zunino, 1987; Abelló *et al*., 1990; Fernández *et al*., 1991). However, although maturity size was larger in the Galician Ría, growth rates were faster, and females attained maturity size faster in the Catalan coast (Abelló *et al*., 1990; Fernández *et al*., 1991). Immature individuals of *L. depurator* did not exhibit growth or 'size' differences between sexes, as these arise later with maturity. Males presented higher growth rates than females, which causes a tendency towards greater relative abundance of males as size increases both in Galicia and in the Mediterranean (Abelló, 1989; Fernández *et al*., 1991).

Crustaceans have hard exoskeletons, often adorned with obvious spines, which should make them excellent

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candidates for shape analysis through geometric morphometrics. Such studies are, however, rare. Rosenberg (1997) studied the shape difference between the major and minor chelipeds of the fiddler crab *Uca pugnax*. The differences observed in the major claw suggested the ability to produce more crushing power and that selection for combat effectiveness might have played an important role in the shape evolution of the major cheliped. The same author extended the study across the genus *Uca* (Rosenberg, 2002). Cadrin & Friedland (1999) and Cadrin (2000) applied box-truss methods (related to geometric morphometrics) to discriminate gender and fishery stocks of the American lobster *Homarus americanus* based on size at maturity (Cadrin, 1995).

The objectives of the current study are (1) to assess whether there is any effect of a digitizing method in geometric morphometric analysis and (2) to analyse shape differences between the carapaces of male and female *L. depurator* from different areas of the Mediterranean coast of Spain.

Materials and methods

A total of 188 *L. depurator* (101 males and 87 females) was collected from the Alborán Sea, Alicante and North Catalonia in the western Mediterranean and frozen (−20 °C) soon after collection. The cephalo-thorax (carapaces from now on) of 125 crabs (63 males and 62 females) were scanned using a calibrated HP Precisionscan 3.1 scanner, and a further 63 crabs (38 males and 25 females) were photographed with a SONY digital camera. Carapace landmarks were identified (see Fig. 1) and used to quantify carapace shape while 'centroid size' (estimated with TpsRel; Rohlf, 2003*b*) was used as a measure of crab 'size'. All crabs studied were right handed to avoid any possible bias of differential handedness in carapace shape. In the study area, most (83.4%) *L. depurator* individuals are right-handed (i.e. the crusher chela is on the right-hand side of the body; Abelló *et al*., 1990).

Figure 1 shows the locations of the 15 carapace landmarks selected (see Rufino, Abelló & Yule, 2004*b*). The first landmark was located centrally on the posterior margin of the carapace. The second landmark was the point of maximum curvature of the posterior carapace margin. Landmarks 3–11 represent the tips of and anterior notch formed by the four anterolateral teeth. Landmarks 12–14 represent the tips of and the notch between the two anterior teeth. Landmark 15 along with landmark 3 delineated the maximum carapace width. Coordinates of the landmarks were measured from the digitized carapace images following Rohlf (2003*a*). Only one half of the carapace was measured, because (1) an increased number of landmarks correspond to a decrease in the statistical power (i.e. a greater number of landmarks correspond to a greater number of shape variables, and therefore, more specimens are needed to equilibrate the matrix for the multivariate analysis) and (2) symmetrical shapes, with corresponding symmetrical landmarks, induce problems in the algebra of the methods and require particular approaches (J. F. Rohlf, pers. comm.).

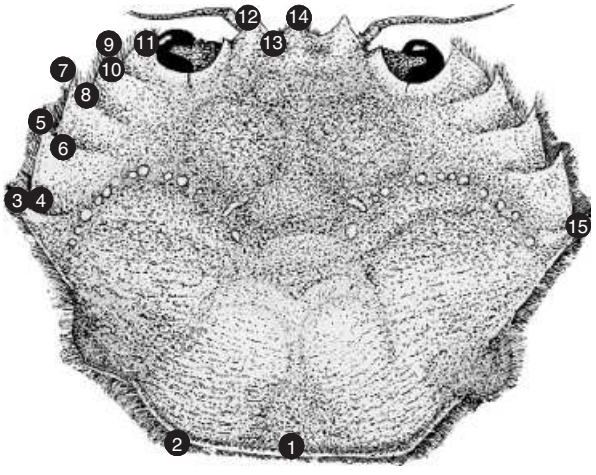


Figure 1 Landmarks selected on the *Liocarcinus depurator* carapace.

Details of the geometric morphometric analysis can be found in Adams, Rohlf & Slice (2003), and references therein. After digitizing, landmark maps were rotated, scaled (to unit centroid size) and translated through a generalized least-squares superimposition (GLS) procedure (generalized Procrustes) to eliminate scale and orientation distortions. A thin-splate spline procedure was used to fit an interpolated function to an average map (consensus configuration) of the carapace shape and derive the uniform and non-uniform (partial warps) components of shape variation. The two uniform components describe differences that affect all parts of the carapace equally (global differences). The non-uniform shape components (partial warps) describe localized departures from the average carapace map.

Relative warps (RWs) analysis (similar to a principal component analysis, using the partial warps scores and the uniform components) was performed to examine the major trends in shape variation. The analysis clearly established a systematic difference in shape resulting from the digitizing technique, showing the camera images to resemble those of smaller size than scanned images equivalent, centroid size.

Posterior carapace width (distance between landmarks 1 and 2), total length (distance between landmarks 1 and 14) and total width (distance between landmarks 3 and 15) were measured on each digitized image. The inter-relationships between these dimensions were then investigated by more traditional techniques and compared between crab gender and geographic location.

Results

Figure 2 shows the results of RW analysis. The first relative warp (RW1) explained 50% and the second (RW2) explained 12% of the total variability. Figure 2 shows that the primordial factor causing differences in shape was centroid size and the first RW of all digitized images was clearly correlated to centroid size (Fig. 2a: $r^2 = 0.56$, $F = 15.254$, $n = 188$, $P < 0.001$). As size is removed from the analysis by the re-scaling procedure, RW1 must indicate a continuous difference in shape from smaller to larger crabs.

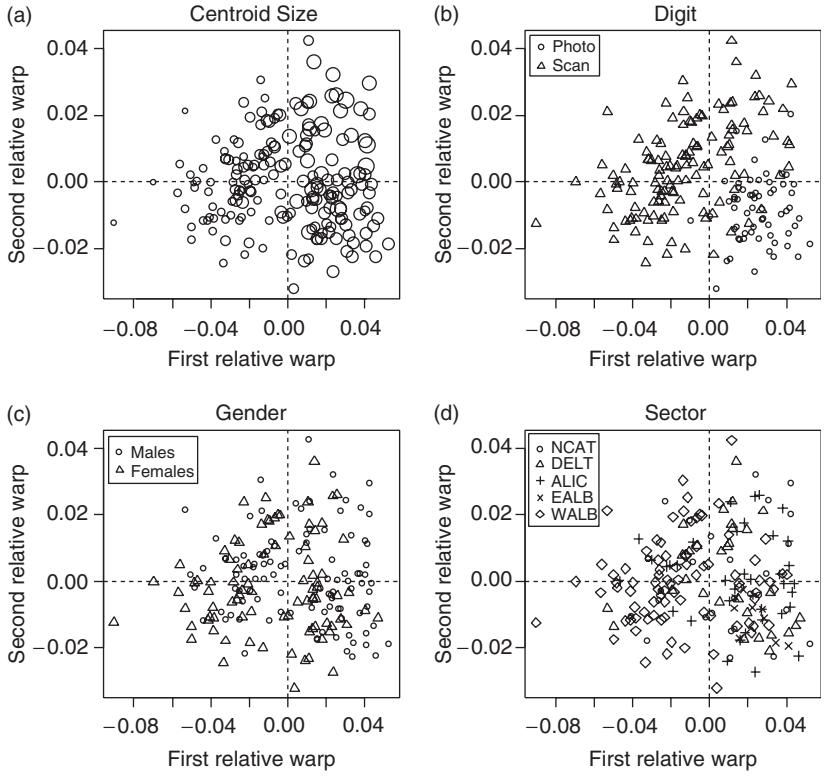


Figure 2 Relative warp scores for all digitized images. Each point is the same in every plot, but uses different symbols to differentiate categories: (a) centroid size is proportional to the circle diameter, (b) digitizing method: photographic camera (circle) or scanner (triangle), (c) gender of the crabs, males (circle) or females (triangle) and (d) geographic location of the samples: NCAT, North Catalonia; DELT, Delta of the Ebro; ALIC, Alicante; EALB, East Alborán Sea; WALB, West Alborán Sea.

The size range of crabs digitized with the scanner was greater than that of the ones digitized with the camera (Fig. 2b), which were only large crabs. Thus, the shape differences between the two digitizing methods are only comparable between large crabs. On the RW1, digitized and scanned images do not differ and the 'size' of the crabs is correctly identified. On RW2, however, the camera and scanned images are clearly separated, with those from the camera consistently more negative on the axis (Fig. 2b). The shape difference between digitizing methods was confirmed to be significant [multiple analysis of variance (MANOVA): Pillai's $\lambda = 0.27$, $F_{(26,161)} = 16.4$, $P < 0.001$], and only the scanner digitized images were consequently used in further analyses. Figure 2c shows that male and female shapes do not separate on either axis, although there was a general tendency for males to spread further along axis 1. The average centroid size for males (♂CS: 0.695 ± 0.02 $\log_{10}(\text{cm}) \pm \text{SE}$) was significantly greater ($T_{[121]} = 2.05$, $P = 0.043$) than that for females (♀CS: $0.640 \pm 0.018 \log_{10}(\text{cm}) \pm \text{SE}$). Ordinations for geographical location (Fig. 2d) also show no great separation on either axis.

Figure 3 highlights the major shape differences (landmark positions) identified between the genders and the southernmost and northernmost sampling locations. Any difference between males and females is minor in comparison with that between northerly and southerly populations. Hence, there appeared to be more landmark differences between populations than between genders, although both involve changes in the relative position of landmarks 15, 2, 3 and 12, which can be summarized by carapace width and length and posterior carapace width.

Table 1a shows the results of the multiple analysis of covariance (MANCOVA) applied to the shape variables (partial warps and uniform components), using the scanned images only. The significant three-way interaction indicates that any differences in shape between sample locations depend on which gender is considered, and no uniform pattern in shape difference is detectable. Sampling location explained 12%, gender 4% and centroid size 35% of the variability in shape, leaving 49% unexplained. Similar results were obtained when the uniform components of shape variation were excluded from the analysis. The three-way interaction is not significant if the analysis is performed using crab images from North Catalonia, Delta and Alicante only, thus excluding samples from the Alborán Sea, which may be under more Atlantic influence. There were then no significant differences between male and female shapes but significant differences between shapes for crabs from Catalonia, Delta and Alicante (Table 1b). Furthermore, significant interaction between location and $\log_{10}$ centroid size indicated a difference in the rate of shape change with growth between the two areas (Table 1b).

Further analysis was carried out by using the dimensions showing the greatest variation in relative landmark position, and considering Delta and North Catalonia as one sector (CAT), to simplify the analysis (see Fig. 3). On average, the measured crabs from the Alborán Sea tended to be smaller than crabs sampled further north (mean posterior carapace

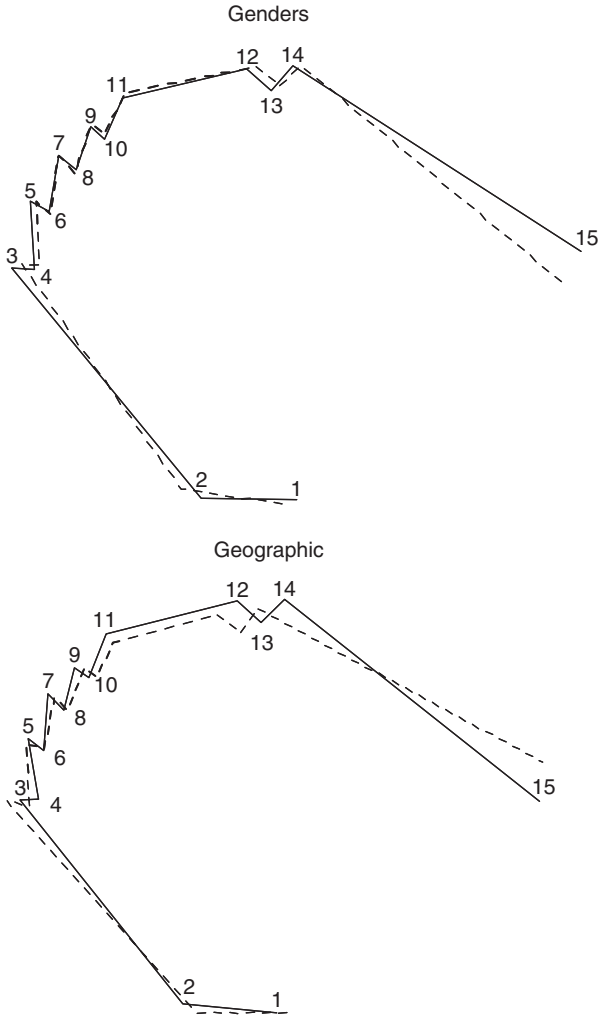


Figure 3 Resistant Procrustes superimposition of the shape differences between males (continuous line) and females (dashed line), and between southernmost (continuous line) and northernmost (dashed line) locations, in each landmark (numbers) (see Fig. 1 for the position of the landmarks). Note that the differences are so small that the values were multiplied by 3 to improve visualization.

width: West Alborán = 0.79 ± 0.04 cm, Alicante = 1.33 ± 0.25 cm and Catalonia = 1.15 ± 0.13 cm), because in the southern locations only small crabs were measured. Furthermore, the relationship between carapace width/length was allometric (power function exponent = 1.047 ± 0.008 and 1.046 ± 0.007 for males and females, respectively). The ratio of carapace width/length was similar in all three areas (mean width/length: West Alborán = 1.27, Alicante = 1.29 and Catalonia = 1.28). The relationships between $\log_{10}$ posterior carapace width and $\log_{10}$ carapace width across gender and location produced a significant three-way interaction (three-way interaction: $F_{[1,49]} = 4.03$, $P = 0.02$), which is clearly shown by the parallel slopes for males and females from the northern crabs in relation to the different slopes for males and females in the south (Fig. 4).

Table 1 Multiple analysis of covariance results showing Pillai's statistic, applied to the partial warps and uniform component of carapace shape in *Liocarcinus depurator*

	Statistic	*F*	d.f.		*P*
(a)					
$\log_{10}$(CS)	0.898	28.491	26	84	<0.001
Gender	0.307	1.433	26	84	0.111
Sector	0.948	1.527	78	258	0.008
Gender × $\log_{10}$(CS)	0.330	1.594	26	84	0.058
Sector × $\log_{10}$(CS)	0.919	1.462	78	258	0.015
Gender × sector	0.837	1.279	78	258	0.080
Gender × sector × $\log_{10}$(CS)	0.890	1.396	78	258	0.028
(b)					
$\log_{10}$(CS)	0.956	17.893	26	21	<0.001
Gender	0.500	0.805	26	21	0.703
Sector	1.464	2.313	52	44	0.003
Gender × $\log_{10}$(CS)	0.497	0.798	26	21	0.711
Sector × $\log_{10}$(CS)	1.371	1.845	52	44	0.02
Gender × sector	1.266	1.459	52	44	0.100
Gender × sector × $\log_{10}$(CS)	1.288	1.531	52	44	0.075

Log centroid size (CS) was used as a covariate: (a) all data [four geographic sectors: North Catalonia (NCAT), Delta of the Ebro (DELT), Alicante (ALIC) and West Alborán Sea (WALB)]; (b) only northern localities: NCAT, DELT and ALIC.

The relationships for the southern and northern crabs were thus re-analysed separately. Table 2 shows that no interaction term was significant for the two northerly locations; hence all slopes were parallel (three-way interaction: $F_{[1]} = 0.84$, $P = 0.364$). However, there was a significant difference in intercept between males and females ($F_{[1,49]} = 4.03$, $P = 0.05$); thus, over the size range examined, females had wider posterior carapaces. There were no significant differences in intercept between the relationships for crabs from either location ($F_{[1,49]} = 0.5$, $P = 0.481$).

In contrast, however, no significant differences between the slopes ($F_{[1,63]} = 0.91$, $P = 0.172$) or the intercepts ($F_{[1,63]} = 0.00$, $P = 0.955$) of the relationship of posterior carapace width versus carapace width were identifiable within the size range measured for the crabs from West Alborán. However, closer examination of Fig. 4 shows that most female crabs over 16 mm in carapace width ($\log_{10} = 0.2$) appear above their male counterparts in a manner very similar to those crabs from the northern areas over the whole size range measured. Females less than 16 mm in carapace width (and several up to 20 mm, $\log_{10} = c.\ 0.3$) are intermingled with their male counterparts.

Figure 5 shows the relationships between *L. depurator* $\log_{10}$ posterior carapace width and the $\log_{10}$ ratio of carapace width to carapace length for each sample location. The crabs from the Alborán Sea showed a much smaller increase in posterior carapace width as the width to length ratio increased than compared with those crabs collected from Alicante and North Catalonia (Table 3). The width of the posterior carapace in short, wide crabs was considerably less for crabs from the south-east than that in equivalent-sized crabs from the more north-easterly areas.

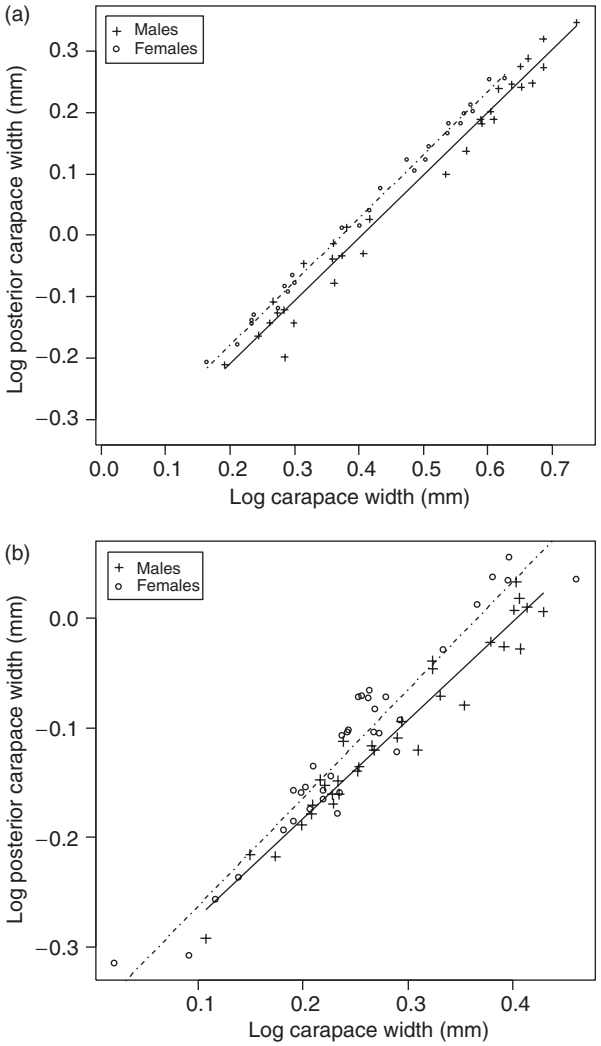


Figure 4 Relationship between $\log_{10}$ posterior carapace width and $\log_{10}$ carapace width of male (+ and continuous line) and female (○ and dashed-dotted line) *Liocarcinus depurator*, in North Catalonia and Alicante (a) and West Alborán Sea (b).

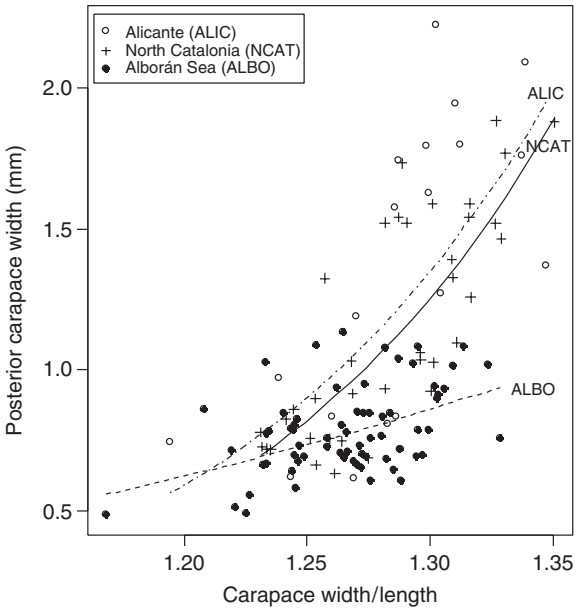


Figure 5 Exponential relationship between posterior carapace width and carapace width/length in male and female *Liocarcinus depurator* sampled from the West Alborán Sea (ALBO) (• and dashed line), Alicante (ALIC) (○ and dashed-dotted line) and North Catalonia (NCAT) (+ and continuous line). The relationships are fitted by least-squares regression.

Figure 5 shows the results of a comparison of the regressions between $\log_{10}$ posterior carapace width and $\log_{10}$ ratio of carapace width to length in *L. depurator* (note that although the regression was performed with log values, the curve shown in the figure corresponds to the back-transformation, i.e. original scale). A highly significant overall regression is indicated ($F_{[1,105]} = 81.16$, $P < 0.001$), and the only significant interaction term showed differences in slope between sampling locations ($F_{[2,105]} = 7.59$, $P = 0.001$). Figure 5 shows that the slopes for both Alicante crabs and those from North Catalonia were similar (no significant difference, Bonferroni $CI_{95\%}$ on the difference $= 0.7 \pm 3.4$) but both slopes were three to four times greater than that for crabs sampled in West Alborán (significant, Bonferroni $CI_{95\%} = 7.1 \pm 3.04$). Within the errors of measurement, there were no significant differences in slope between males and females irrespective of sample location ($F_{[1,105]} = 1.24$, $P = 0.267$), although the slope for females was always less than that for males.

Discussion

Aldrich (1993) suggested that crustacean taxonomy and even shape differences between local populations of the same species can be reliably examined by using a global approach to shape analysis through geometric morphometrics. The present study shows that the application of geometric morphometric analysis to the portunid crab *L. depurator* carapace shape provides results that are interpretable in terms of its population biology even though the differences may be small. The inability of multivariate statistics to separate the populations probabilistically results from significance in a high-order interaction term which are notoriously unstable under MANOVA (see Table 1). The interaction undoubtedly arises because of the relative growth differences between males and females from the southerly populations and those in the north (see Fig. 4).

It is clear from the current study that the digitizing method considerably influences the results of the morphometric analysis. The digital camera produced images that probably resulted in more variability for a given crab's size and had a distinct systematic departure from those obtained from the scanner. As in all scientific endeavours, it is essential to find the appropriate tool with the correct resolution to tackle the problem. Furthermore, it is abundantly clear that images captured by one device are unlikely to match those captured with another; hence, mixing tools in

Table 2 Results of an analysis of variance of $\log_{10}$ carapace width and $\log_{10}$ posterior carapace width of *Liocarcinus depurator* in the northern Mediterranean coast of Spain [Alicante and Catalonia (a) and (b)] and in the south [(Alborán (c) and (d)]

(a) *Slopes and intercepts for northern crabs*

	Location					
	Alicante		Catalonia		Mean	
	Slope ± SE	Intercept ± SE	Slope ± SE	Intercept ± SE	Slope	Intercept
Males	1.066 ± 0.045	−0.436 ± 0.025	0.999 ± 0.039	−0.404 ± 0.019	1.033	−0.42
Females	1.027 ± 0.021	−0.381 ± 0.009	1.032 ± 0.023	−0.386 ± 0.1	1.03	−0.384
Mean	1.047	−0.409	1.016	−0.395	1.032	−0.402

(b) *Analysis of variance table for northern crabs*

Variable	d.f.	Sum of squares	Mean adj. SS	*F*	*P*
Gender	1	0.00962	0.00190	4.03	0.050
Sector	1	0.03373	0.00024	0.50	0.481
$\log_{10}$width	1	1.42916	1.32700	2822.39	<0.001
Gender × sector	1	0.00003	0.00049	1.04	0.313
Gender × $\log_{10}$width	1	0.00000	0.00000	0.01	0.940
Sector × $\log_{10}$width	1	0.00049	0.00030	0.63	0.432
Gender × sector × $\log_{10}$width	1	0.00039	0.00039	0.84	0.364
Error	49	0.02304	0.00047		

(c) *Slopes and intercepts of Alborán crabs*

Gender	Slope ± SE	Intercept ± SE
Males	0.988 ± 0.050	−0.362 ± 0.013
Females	0.899 ± 0.039	−0.363 ± 0.012
Mean	0.944	−0.363

(d) *Analysis of variance table for the Alborán crabs*

Variable	d.f.	Sum of squares	Mean adj. SS	*F*	*P*
Gender	1	0.00228	0.00000	0.00	0.955
Log_{10} width	1	0.43993	0.43581	862.20	<0.001
Gender × $\log_{10}$ width	1	0.00096	0.00096	1.91	0.172
Error	63	0.03184	0.00051		

Sums of squares are adjusted for entry order.

a single investigation will undoubtedly compromise the result. A previous study has already highlighted the problem (M. Corti, pers. comm.) where image capture of the same specimens by two different digital cameras gave quite contrasting results when analysed. The proofing of novel or little documented techniques should be uppermost in any investigator's mind before data collection, let alone analysis.

Unlike in the current study, Abelló *et al.* (1990) failed to detect sexual dimorphism in *L. depurator* carapace shape through non-geometric morphometrics. The current study and that of Rufino, Abelló & Jones (2004*a*), however, showed that female *L. depurator* have a wider posterior carapace base than the males using both geometric and non-geometric morphometrics. A wider posterior carapace may result in a greater relative volume for gonad development within the carapace (see Hartnoll, 1982, 1985). The present results not only demonstrate that females have wider posterior carapace bases than males but that this measure is associated with both males and females as the width to length ratio of the carapace increases. The relationship between carapace width and length is allometric, with crabs tending to become wider rather than longer as they grow.

Geometric morphometrics helped to identify the areas of the carapace showing the greatest variability across sample locations, and hence prompted the choice of linear dimension analysis used to examine the underlying differences identified through the multivariate analysis of the shape variables.

The relationship between the width of the posterior carapace and the ratio of carapace length/width differed significantly, with a much lower rate of posterior carapace increase with ratio in the southern samples in comparison with those from further north. Although the crabs measured from the Alborán Sea were smaller than those of the northern areas, there was still a considerable overlap in the size range of crabs between the three samples. The width/length ratio increased as the crabs became larger and hence the broadening of the posterior carapace is much less pronounced with size in crabs from West Alborán. Indeed, it would appear that southern crabs with smaller width/length ratios actually have wider posterior carapaces on average than their counterparts in the north.

The Alborán Sea, off the south-western Mediterranean coast of Spain, is heavily influenced by the influx of Atlantic

Table 3 Results of an analysis of variance using $\log_{10}$ ratio of carapace width to length as a covariate, comparing the relationships between $\log_{10}$ posterior carapace width and $\log_{10}$ ratio for male and female *Liocarcinus depurator* from three areas of the Mediterranean coast of Spain

(a) *Slopes* ± *SE*

Location	Males	Females	Average
West Alborán	3.5 ± 1.7	1.8 ± 1.2	4.4 ± 1.2
Alicante	10.4 ± 2.3	9.8 ± 2.2	10.1 ± 1.5
North Catalonia	11.3 + 1.7	7.4 ± 1.8	9.4 ± 1.3

(b) *Analysis of variance table*

Variable	d.f.	Sum of squares	Adj. mean SS	*F*	*P*
Gender	1	0.02356	0.01185	1.35	0.248
Sector	2	0.91768	0.04380	4.98	0.009
$\log_{10}$width/length	1	0.62676	0.71315	81.16	<0.001
Gender × sector	2	0.00178	0.00269	0.31	0.737
Gender × $\log_{10}$width/length	1	0.03372	0.01094	1.24	0.267
Sector × $\log_{10}$width/length	2	0.13959	0.06667	7.59	0.001
Gender × sector × $\log_{10}$width/length	2	0.00742	0.00371	0.42	0.657
Error	105	0.92268	0.00879		

Sums of squares are adjusted for entry order.

water entering the Mediterranean through the straits of Gibraltar on the upper levels of the water column, and of the Mediterranean waters exiting the basin into the Atlantic Ocean through deeper levels (Hopkins, 1985; Millot, 1999). The stable occurrence of the Almería – Orán hydrographical front (Font *et al*., 1998; Millot, 1999) delimitates the Atlantic waters from the typically Mediterranean waters. These hydrographical structures affecting the water dynamics have been identified as one, if not the most important, factor contributing to delimitate faunistic regions, as well as isolated or semi-isolated populations of marine animals (Maldonado & Uriz, 1995; Medel & López-González, 1996; De Grave & Diaz, 2001; Abelló *et al*., 2002), as community structure in the Alborán Sea can exhibit characteristics more closely resembling Atlantic assemblages of the same species than those found in the rest of the Mediterranean. The waters off Catalonia are under the influence of the strong Liguro-Provençal-Catalan current, which flows along the southern coasts of Europe delimitating coastal and offshore Mediterranean warmer, more saline waters (e.g. Salat, 1996). Thus, the Atlantic influence is much less pronounced on the northern Spanish coast than in the south. Given the high influx rate of Atlantic water, it is feasible for epiplanktonic larvae from populations outside the Mediterranean to gain access and potentially establish themselves in the Alborán Sea and areas further away under the direct influence of the incoming Atlantic water, such as Algeria – Tunisia, the Balearic Islands or Sicily. It seems much less likely that such incoming larvae could mix with populations in Catalonia or Alicante, because of the prevailing oceanographical conditions. The Alicante area shows a sporadic direct influence of Atlantic waters, but shows more permanent true Mediterranean influence from northern currents. In general, morphological variability between populations is attributed to genetics through isolation by frontal structures, temperature refuges (glacial relict species in the deep Mediterranean Sea) or different environmental conditions.

Among crustacean species occurring in the Mediterranean as well as in the Atlantic, morphometric analyses have been applied by De Grave & Diaz (2001), who compared the morphology (linear dimension analysis) of Mediterranean and Atlantic populations of the crangonid shrimp *Pontophilus norvegicus*. They found that telson length, antennal scale width and the presence of a tubercle on the dorsal carina and carapace showed discriminating power for separating populations from the Atlantic and the Mediterranean. This species is considered a glacial relict in the deep Mediterranean, and its Mediterranean and Atlantic populations have become completely isolated (Abelló & Valladares, 1985).

It is likely that the Alborán Sea population of *L. depurator*, given the oceanographic characteristics of the area, is largely recruited from the Atlantic, as well as from its own population, rather than from more northern Mediterranean populations, separated by the Almería–Oran front. In Atlantic waters, marine species tend to reach larger sizes. It is possible that the slower increase in posterior carapace width with carapace width/length ratio indicates a later onset of maturity with size in the Alborán sea. Figure 4a shows no evidence of a major maturity shape change for crabs in the northern areas yet clear evidence of shape change at a given size range for crabs from the Alborán Sea (Fig. 4b). The slower growth rates observed in Atlantic species possibly accompany a retarded rate of maturity, which is reflected in the Alborán crabs but not in the crabs from the Catalan and Alicante coasts. The paucity of small specimens (<16 mm carapace width) from the northern samples and large specimens from the Alborán area samples (Fig. 4a) does not allow further conclusions from the current data.

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